大作业: 上机报告

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目录

1 问题描述：Stokes方程与交错网格上的MAC格式

1.1 Stokes方程

考虑Stokes方程

$$\begin{cases} -\Delta \vec{\mathbf{u}} + \nabla p = \vec{\mathbf{F}}, & (x,y) \in (0,1)\times(0,1) \\ div\ \vec{\mathbf{u}} = 0, & (x,y) \in (0,1)\times(0,1) \end{cases}$$

边界条件为

$$\begin{aligned} &\frac{\partial u}{\partial \vec{\mathbf{n}}} = b, \quad y = 0, \qquad \frac{\partial u}{\partial \vec{\mathbf{n}}} = b, \quad y = 1 \\ &\frac{\partial v}{\partial \vec{\mathbf{n}}} = l, \quad x = 0, \qquad \frac{\partial v}{\partial \vec{\mathbf{n}}} = r, \quad x = 1 \\ &u = 0, \quad x = 0,1, \qquad v = 0, \quad y = 0,1 \end{aligned}$$

其中, $\vec{\mathbf{u}} = (u,v)$为速度, p为压力, $\vec{\mathbf{F}} = (f,g)$为外力, $\vec{\mathbf{n}}$为外法向向量

1.2 MAC格式离散

利用交错网格上的MAC格式离散Stokes方程, 记$h = \frac{1}{N}$

$$U_{(N+1)\times N} = (U_{ij}), \qquad U_{ij} \longrightarrow u(x_i, y_j), \qquad x_i = (i-1)h, y_i = (j-\frac{1}{2})h$$

$$V_{N\times(N+1)} = (V_{ij}), \qquad V_{ij} \longrightarrow v(x_i, y_j), \qquad x_i = (i-\frac{1}{2})h, y_i = (j-1)h$$

$$P_{N\times N} = (P_{ij}), \qquad P_{ij} \longrightarrow p(x_i, y_j), \qquad x_i = (i-\frac{1}{2})h, y_i = (j-\frac{1}{2})h$$

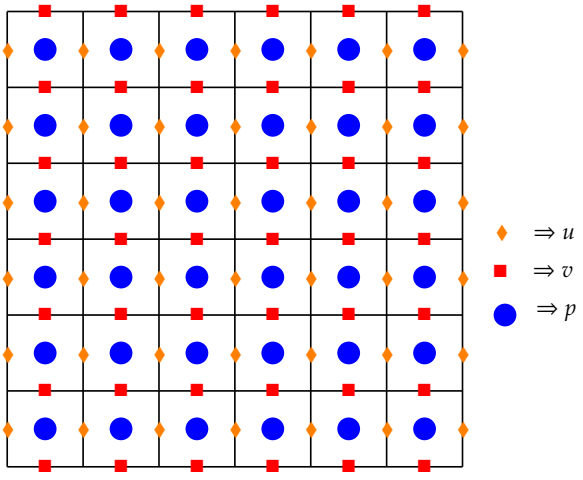

图 1: MAC离散(此处图片来自讲义)

可得如下方程

$if \qquad 2 \le i \le N, 2 \le j \le N-1, \qquad x_i = (i-1)h, y_i = (j-\frac{1}{2})h$

$$-\frac{U_{i-1,j} - 2U_{i,j} + U_{i+1,j}}{h^2} - \frac{U_{i,j-1} - 2U_{i,j} + U_{i,j+1}}{h^2} + \frac{P_{i,j} - P_{i-1,j}}{h} = f(x_i, y_j)$$

$if \quad 2 \le i \le N, \qquad x_i = (i-1)h$

$$-\frac{U_{i-1,1} - 2U_{i,1} + U_{i+1,1}}{h^2} - \frac{U_{i,2} - U_{i,1}}{h^2} - \frac{b(x_i)}{h} + \frac{P_{i,1} - P_{i-1,1}}{h} = f(x_i, \frac{1}{2}h)$$

$$-\frac{U_{i-1,N} - 2U_{i,N} + U_{i+1,N}}{h^2} - \frac{U_{i,N} - U_{i,N-1}}{h^2} - \frac{t(x_i)}{h} + \frac{P_{i,1} - P_{i-1,1}}{h} = f(x_i, (N - \frac{1}{2})h)$$

$if \quad i = 1 or N+1$

$$U_{i,j} = 0 \qquad \forall j$$

$if \quad 2 \le j \le N, 2 \le i \le N-1, \qquad x_i = (i - \frac{1}{2})h, y_i = (j-1)h$

$$-\frac{V_{i-1,j} - 2V_{i,j} + V_{i+1,j}}{h^2} - \frac{V_{i,j-1} - 2V_{i,j} + V_{i,j+1}}{h^2} + \frac{P_{i,j} - P_{i,j-1}}{h} = g(x_i, y_j)$$

$if \quad 2 \le j \le N, \qquad y_j = (j-1)h$

$$-\frac{V_{1,i-1} - 2V_{1,i} + V_{1,i+1}}{h^2} - \frac{V_{2,i} - V_{1,i}}{h^2} - \frac{l(x_i)}{h} + \frac{P_{1,i} - P_{1,i-1}}{h} = g(\frac{1}{2}h, y_j)$$

$$-\frac{V_{N,i-1} - 2V_{N,i} + V_{N,i+1}}{h^2} - \frac{V_{N,i} - V_{N-1,i}}{h^2} - \frac{r(x_i)}{h} + \frac{P_{1,i} - P_{1,i-1}}{h} = g((N - \frac{1}{2})h, y_j)$$

$if \quad j = 1 or N+1$

$$V_{i,j} = 0 \qquad \forall j$$

$if \quad 1 \le i \le N, 1 \le j \le N$

$$\frac{U_{i+1,j} - U_{i,j}}{h} + \frac{V_{i,j+1} - V_{i,j}}{h} = 0$$

将b, t, l, r移动至RHS, 并对系数做处理使得满足对称关系, 做离散

$$F_{(N+1)\times N} = (F_{ij}), \qquad F_{ij} \longrightarrow f(x_i, y_j)(\text{边界处考虑} b, t), \qquad x_i = (i-1)h, y_i = (j - \frac{1}{2})h$$

$$G_{N\times(N+1)} = (G_{ij}), \qquad G_{ij} \longrightarrow g(x_i, y_j)(\text{边界处考虑} l, r), \qquad x_i = (i - \frac{1}{2})h, y_i = (j-1)h$$

最终得到线性方程如下

$if \quad 2 \le i \le N, 2 \le j \le N-1$

$$-\frac{U_{i-1,j} - 2U_{i,j} + U_{i+1,j}}{h^2} - \frac{U_{i,j-1} - 2U_{i,j} + U_{i,j+1}}{h^2} + \frac{P_{i,j} - P_{i-1,j}}{h} = F_{i,j}$$

$if \quad 2 \le i \le N$

$$-\frac{U_{i-1,1} - 2U_{i,1} + U_{i+1,1}}{h^2} - \frac{U_{i,2} - U_{i,1}}{h^2} + \frac{P_{i,1} - P_{i-1,1}}{h} = F_{i,1}$$

$$-\frac{U_{i-1,N} - 2U_{i,N} + U_{i+1,N}}{h^2} - \frac{U_{i,N} - U_{i,N-1}}{h^2} + \frac{P_{i,1} - P_{i-1,1}}{h} = F_{i,N}$$

$if \quad i = 1 or N+1$

$$U_{i,j} = 0 \qquad \forall j$$

$if \quad 2 \le j \le N, 2 \le i \le N-1$

$$-\frac{V_{i-1,j} - 2V_{i,j} + V_{i+1,j}}{h^2} - \frac{V_{i,j-1} - 2V_{i,j} + V_{i,j+1}}{h^2} + \frac{P_{i,j} - P_{i,j-1}}{h} = G_{i,j}$$

$if \quad 2 \le j \le N$

$$-\frac{V_{1,i-1} - 2V_{1,i} + V_{1,i+1}}{h^2} - \frac{V_{2,i} - V_{1,i}}{h^2} + \frac{P_{1,i} - P_{1,i-1}}{h} = G_{1,j}$$

$$-\frac{V_{N,i-1}-2V_{N,i}+V_{N,i+1}}{h^2}-\frac{V_{N,i}-V_{N-1,i}}{h^2}+\frac{P_{1,i}-P_{1,i-1}}{h}=G_{N,j}$$

$if \qquad j=1 or N+1$

$$V_{i,j}=0 \qquad \forall j$$

$if \qquad 1\le i\le N, 1\le j\le N$

$$-\frac{U_{i+1,j}-U_{i,j}}{h}-\frac{V_{i,j+1}-V_{i,j}}{h}=0$$

其可以写成格式

$$\begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix}\begin{pmatrix} X \\ P \end{pmatrix}=\begin{pmatrix} Y \\ 0 \end{pmatrix}$$

其中, X是U,V的向量化, Y是F,G的向量化, 通过改变最后一式的符号, 保证了对称性

1.3 数值算例

在$\Omega=(0,1)\times(0,1)$上

$$f(x,y)=-4\pi^2(2cos(2\pi x)-1)sin(2\pi y)+x^2$$

$$g(x,y)=4\pi^2(2cos(2\pi y)-1)sin(2\pi x)$$

$$b(x)=-2\pi(1-cos(2\pi x))$$

$$t(x)=2\pi(1-cos(2\pi x))$$

$$l(y)=2\pi(1-cos(2\pi y))$$

$$r(y)=-2\pi(1-cos(2\pi y))$$

真解为

$$u(x,y)=(1-cos(2\pi x)sin(2\pi y)$$

$$v(x,y)=-(1-cos(2\pi y))sin(2\pi x)$$

$$p(x,y)=\frac{x^3}{3}-\frac{1}{12}$$

2 P1: 使用 DGS 磨光子

2.1 数值方法

2.1.1 V-cycle多重网格法

用V-cycle求解方程

$$AV = F, \qquad A \in \mathbb{R}^{N\times N}, V, F \in \mathbb{R}^{N\times 1}$$

$$A_h = A, V_h = V, F_h = F, \qquad h = \frac{1}{N}, \qquad N = 2^n$$

step1: 用DGS迭代法, 以$V_h^{(0)} = 0$为初值, 做v_1次迭代得$V_h^{(v_1)}$

$$R_h = F_h - A_h V_h^{v_1}$$

step2: 令

$$A_{2h} = I_h^{2h} A_h I_{2h}^h$$

其中I_h^{2h}为从N到$\frac{1}{2}N$的限制算子, I_{2h}^h为从$\frac{1}{2}N$到N的提升算子
特别地, 对于原问题, 我们取A_{2h}为$\frac{1}{2}N$时的 MAC 离散化所得(此处参考**提示5**)
令

$$R_{2h} = I_h^{2h} R_h$$

解方程

$$A_{2h} V_{2h} = R_{2h}$$

以$V_{2h}^{(0)}$为初值(第1个 V-cylce 中取为0), 做v_1次 DGS 迭代, 得$V_{2h}^{(v1)}$以及

$$R_{2h} = F_{2h} - A_{2h} V_{2h}^{(v_1)}$$

step(n-l): 得到$V_{2^{n-l-1}h}^{(v1)}$, 以及

$$R_{2^{n-l-1}h} = F_{2^{n-l-1}h} - A_{2^{n-l-1}h} V_{2^{n-l-1}h}^{(v_1)}$$

step(n-l+1): 令

$$A_{2^{n-l}h} = I_{2^{n-l-1}h}^{2^{n-l}h} A_{2^{n-l-1}h} I_{2^{n-l-1}h}^{2^{n-l}h}$$

并令

$$F_{2^{n-l}h} = I_{2^{n-l-1}h}^{2^{n-l}h} R_{2^{n-l-1}h}$$

直接求解

$$A_{2^{n-l}h} V_{2^{n-l}h} = F_{2^{n-l}h}$$

得$V_{2^{n-l}h}$

step(n-l)': 计算

$$V_{2^{n-l-1}h}^{(v_1+1)} = V_{2^{n-l-1}h}^{(v_1)} + I_{2^{n-l}h}^{2^{n-l-1}h} V_{2^{n-l}h}$$

以$V_{2^{n-l-1}h}^{(v_1+1)}$为初值, 求解

$$A_{2^{n-l-1}h} V_{2^{n-l-1}h} = F_{2^{n-l-1}h}$$

做v_2次G-S迭代, 得$V_{2^{n-l-1}h}^{(v_1+v_2+1)}$

step1': 计算

$$V_h^{(v_1+1)} = V_h^{(v_1)} + I_{2h}^h V_{2h}$$

以$V_h^{(v_1+1)}$为初值, 求解

$$A_h V_h = F_h$$

做v_2次DGS迭代, 得$V_h^{(v_1+v_2+1)}$

计算

$$r_h = F_h - A_h V_h^{(v_1+v_2+1)}, \qquad r_0 = F_h - A_h V_h^{(0)}$$

停止条件为

$$\frac{\|r_h\|_2}{\|r_0\|_2} < 10^{-8}$$

如满足条件, 停止迭代, $V_h^{(v_1+v_2+1)}$为近似解

否则回到step1, 以$V_h^{(0)} = V_h^{(v_1+v_2+1)}$为初值做 V-cylce

2.1.2 DGS迭代法

用 DGS 方法对方程

$$\begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} X \\ P \end{pmatrix} = \begin{pmatrix} Y \\ 0 \end{pmatrix}$$

做一次迭代

step1: 先对X做一次 Gauss-Seidel 迭代

$$R \longleftarrow Y - (AX + BP)$$

$$R \longrightarrow RF, RG$$

用RF, RG, 分别进行逐个更新, 得到EU, EV, 其中, 对U按$j = 1 : N, i = 1 : N + 1$顺序更新, 对V按$j = 1 : N, i = 1 : N + 1$顺序更新, 具体地从方程导出迭代格式

$j = 1, i = 1$

$$U_{1,1} = 0 \Longrightarrow EU_{1,1} = RF_{1,1}$$

$j = 1, i = 2 : N$

$$-\frac{U_{i-1,1} - 2U_{i,1} + U_{i+1,1}}{h^2} - \frac{U_{i,2} - U_{i,1}}{h^2} + \frac{P_{i,1} - P_{i-1,1}}{h} = F(i,1)$$

$$\Longrightarrow -\frac{EU_{i-1,1} - 2EU_{i,1}}{h^2} - \frac{-EU_{i,1}}{h^2} = RF(i,1)$$

$$\Longrightarrow EU_{i,1} = \frac{1}{3} EU_{i-1,1} + \frac{1}{3} h^2 RF_{i,1}$$

$j = 1, i = N + 1,$

$$U_{N+1,1} = 0 \Longrightarrow EU_{N+1,1} = RF_{N+1,1}$$

$j = 2 : N - 1, i = 1$

$$U_{1,j} = 0 \Longrightarrow EU_{1,j} = RF_{1,j}$$

$j = 2 : N - 1, i = 2 : N$

$$-\frac{U_{i-1,j} - 2U_{i,j} + U_{i+1,j}}{h^2} - \frac{U_{i,j-1} - 2U_{i,j} + U_{i,j+1}}{h^2} + \frac{P_{i,j} - P_{i-1,j}}{h} = F_{i,j}$$

$$\Longrightarrow -\frac{EU_{i-1,j} - 2EU_{i,j}}{h^2} - \frac{EU_{i,j-1} - 2EU_{i,j}}{h^2} = RF_{i,j}$$

$$\Longrightarrow EU_{i,j} = \frac{1}{4}EU_{i,j-1} + \frac{1}{4}EU_{i-1,j} + \frac{1}{4}h^2 RF_{i,j}$$

$j = 2 : N-1, i = N+1$

$$U_{N+1,j} = 0 \Longrightarrow EU_{N+1,j} = RF_{N+1,j}$$

$j = N, i = 1$

$$U_{i,N} = 0 \Longrightarrow EU_{i,N} = RF_{i,N}$$

$j = N, i = 2 : N$

$$-\frac{U_{i-1,N} - 2U_{i,N} + U_{i+1,N}}{h^2} - \frac{U_{i,N} - U_{i,N-1}}{h^2} + \frac{P_{i,1} - P_{i-1,1}}{h} = F_{i,N}$$

$$\Longrightarrow -\frac{EU_{i-1,N} - 2EU_{i,N}}{h^2} - \frac{EU_{i,N} - EU_{i,N-1}}{h^2} = RF_{i,N}$$

$$\Longrightarrow EU_{i,N} = \frac{1}{3}EU_{i+1,N} + \frac{1}{3}EU_{i,N-1} + \frac{1}{3}h^2 RF_{i,N}$$

$j = 1, i = N+1,$

$$U_{N+1,N} = 0 \Longrightarrow EU_{N+1,N} = RF_{N+1,N}$$

再计算

$$U_{half} = U + EU$$

对V, 同理地进行更新, 得V_{half}

step2: 再更新压力

总体地, 采用内部单元⟶顶边⟶底边⟶左边⟶右边⟶顶点的顺序更新压力

这里, 与讲义中不同的是, 将待求解的方程写做

$$\begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} X \\ P \end{pmatrix} = \begin{pmatrix} Y \\ Z \neq 0 \end{pmatrix}$$

计算梯度方程残量时, 将Z计入(此处参考**提示3**)

(内部单元)$i : 2 : N-1, j = 2 : N-1$

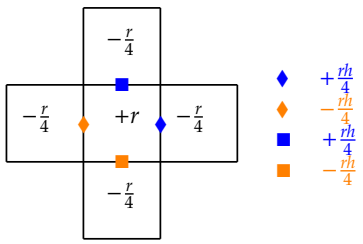

图 2: 内部单元(此处图片来自讲义)

计算散度方程残量

$$r_{i,j} = -\frac{U^{half}_{i+1,j} - U^{half}_{i,j}}{h} - \frac{V^{half}_{i,j+1} - V^{half}_{i,j}}{h} - Z_{i,j}$$

$$\delta = \frac{r_{i,j}h}{4}$$

更新内部单元速度

$$U^{half}_{i,j} = U^{half}_{i,j} - \delta, \qquad U^{half}_{i+1,j} = U^{half}_{i+1,j} + \delta$$

$$V_{i,j}^{half} = V_{i,j}^{half} - \delta, \qquad V_{i,j+1}^{half} = V_{i,j+1}^{half} + \delta$$

更新内部单元压力

$$P_{i,j} = P_{i,j} + r \qquad P_{i+1,j} = P_{i+1,j} - \frac{1}{4}r, \qquad P_{i-1,j} = P_{i-1,j} - \frac{1}{4}r$$

$$P_{i,j+1} = P_{i,j+1} - \frac{1}{4}r, \qquad P_{i,j-1} = P_{i,j-1} - \frac{1}{4}r$$

(顶边)$i = 1 : N$

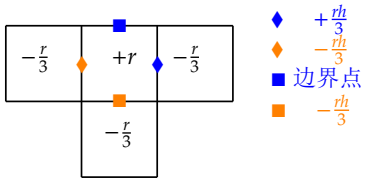

图 3: 边界单元(此处图片来自讲义)

计算散度方程残量

$$r_{i,1} = -\frac{U_{i+1,1}^{half} - U_{i,1}^{half}}{h} - \frac{V_{i,2}^{half} - V_{i,1}^{half}}{h} - Z_{i,1}$$

$$\delta = \frac{r_{i,1}h}{3}$$

更新内部单元速度

$$U_{i,1}^{half} = U_{i,1}^{half} - \delta, \qquad U_{i+1,1}^{half} = U_{i+1,1}^{half} + \delta$$

$$V_{i,2}^{half} = V_{i,2}^{half} + \delta$$

更新内部单元压力

$$P_{i,1} = P_{i,1} + r \qquad P_{i+1,1} = P_{i+1,1} - \frac{1}{3}r$$

$$P_{i-1,1} = P_{i-1,1} - \frac{1}{3}r, \qquad P_{i,j+1} = P_{i,j+1} - \frac{1}{3}r$$

对底边,左边, 右边, 同理地进行更新

(顶点)$i = 1, j = 1$

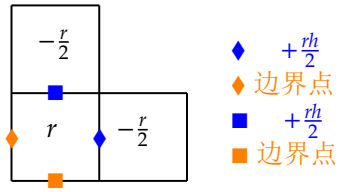

图 4: 顶点单元(此处图片来自讲义)

计算散度方程残量

$$r_{1,1} = -\frac{U_{2,1}^{half} - U_{1,1}^{half}}{h} - \frac{V_{1,2}^{half} - V_{1,1}^{half}}{h} - Z_{1,1}$$

$$\delta = \frac{r_{1,1}h}{2}$$

更新内部单元速度

$$U_{2,1}^{half} = U_{2,1}^{half} + \delta \qquad V_{1,2}^{half} = V_{1,2}^{half} + \delta$$

更新内部单元压力

$$P_{1,1} = P_{1,1} + r \qquad P_{2,1} = P_{2,1} - \frac{1}{2}r, \qquad P_{1,2} = P_{1,2} - \frac{1}{2}r$$

对另外三个顶点, 同理地进行更新

2.1.3 限制算子和提升算子

采用如图所示的限制算子和提升算子

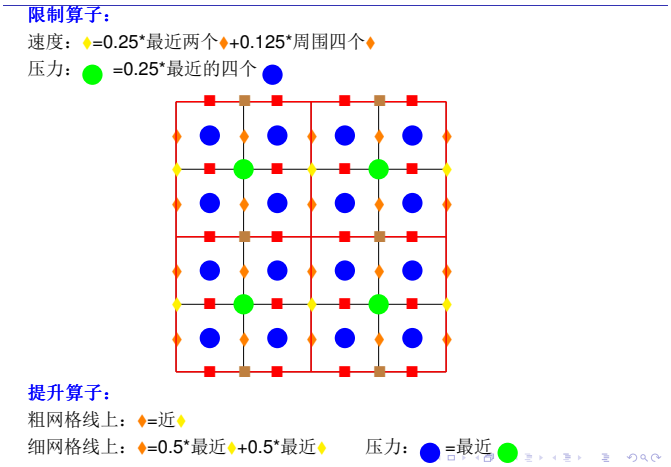

图 5: 限制算子和提升算子(此处图片来自讲义)

2.2 具体算法实现

表 1: 辅助函数

文件	作用
p1:/initialize.m	初始化F, G及计算真解
p1:/left_multiply_matrix.m	对矩阵U, V的向量形式左乘$[A, B; B^T, 0]$后化回矩阵形式
p1:/limit.m	对F, G, Z应用限制算子, 保持其矩阵形式
p1:/lift.m	对U, V, P应用提升算子, 保持其矩阵形式

Algorithm 1 V-cycle 多重网格法(p1:/V_cycle.m)

$U, V, P = 0, iter1 = log_2(N) - log_2(L)$
while $r_h/r_0 > 10^{-8}$ **and** $iter < MAXITER$ **do**
  $iter = iter + 1$
  for $i = 1 : iter1$ **do**
    for $k = 1 : v_1$ **do**
      $[U, V, P]$ =**DGS**(U, V, P, F, G, Z, N)
    end for
    $store\ U, V, P, F, G, Z, N$
    $[U1, V1, P1]$ =**left_multiply_matrix**(U, V, P, N)
    $[F, G, Z]$ =**limit**$(F - U1, G - V1, Z - P1, N)$
    $N = N/2$
  end for
  $solve\ U, V, P, F, G, Z \longrightarrow U, V, P$
  for $i = iter1 : -1 : 1$ **do**
    $[U, V, P]$ =**lift**(U, V, P, N)
    $N = N * 2$
    $U, V, P = store + U, V, P$
    for $k = 1 : v_2$ **do**
      $[U, V, P]$ =**DGS**$(U, V, P, Fstore, Gstore, Zstore, N)$
    end for
    if $i == 1$ **then**
      $store\ U, V, P$
    end if
  end for
end while

Algorithm 2 DGS迭代法(p1:/DGS.m)

$[U1, V1, \ ]$ =**left_multiply_matrix**(U, V, P, N)
$RF = F - U1, RV = G - V1$
$Gauss - Seidel \longrightarrow EU, EV$
$U_{half} = U + EU, V_{half} = V + EV$
for $i = 2 : N - 1$ **do**
  for $j = 2 : N - 1$ **do**
    $update\ P(i, j)$
  end for
end for
for $i = 2 : N - 1$ **do**
  $update\ P(i, 1), P(i, N)$
end for
for $j = 2 : N - 1$ **do**
  $update\ P(1, j), P(N, j)$
end for
$updateP(1, 1), P(1, N), P(N, 1), P(N, N)$

2.3 数值结果与分析

分别取$N = 64, 128, 256, 512, 1024, 2048,$ 以 DGS 为磨光子, 用基于 V-cycle 的多重网格方法求解离散问题, 停机标准为$\|r_h\|_2/\|r_0\|_2 < 10^{-8}$, 对不同的$v_1, v_2, L,$ 比较 V-cycle 的次数和 CPU 时间, 并计算误差

$$e_N = h\left(\sum_{j=1}^{N}\sum_{i=1}^{N-1}\left|u_{i,j-\frac{1}{2}} - u(x_i, y_{j-\frac{1}{2}})\right|^2 + \sum_{j=1}^{N-1}\sum_{i=1}^{N}\left|v_{i-\frac{1}{2},j} - v(x_{i-\frac{1}{2}}, y_j)\right|^2\right)$$

其中$x_i = i\frac{1}{N}, y_j = j\frac{1}{N}$,　$u(i, j), v(i, j)$分别为计算结果在(x_i, y_j)处的取值
结果如下

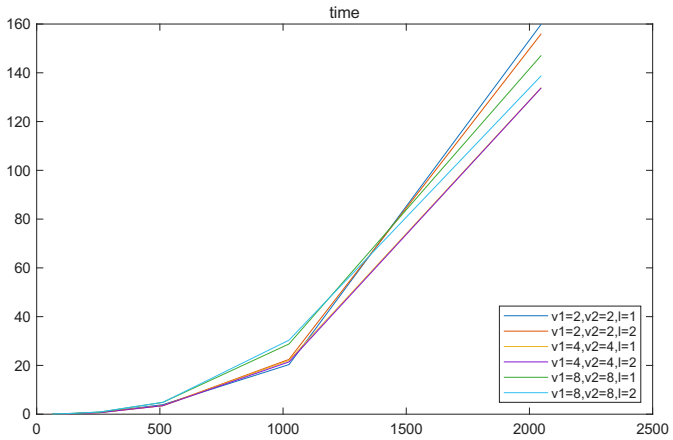

图 6: 时间随N的变化

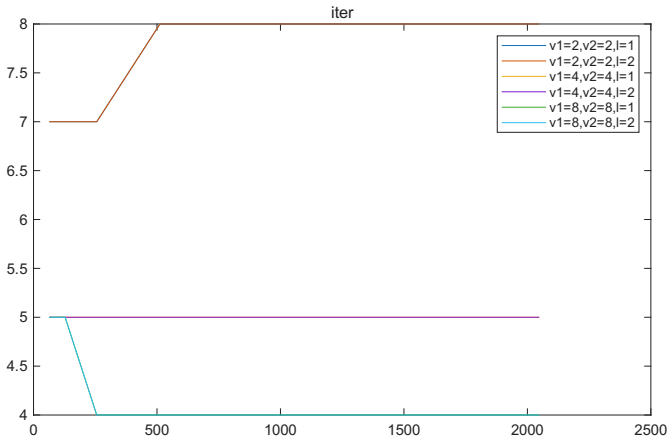

图 7: V-cycle 次数随N的变化

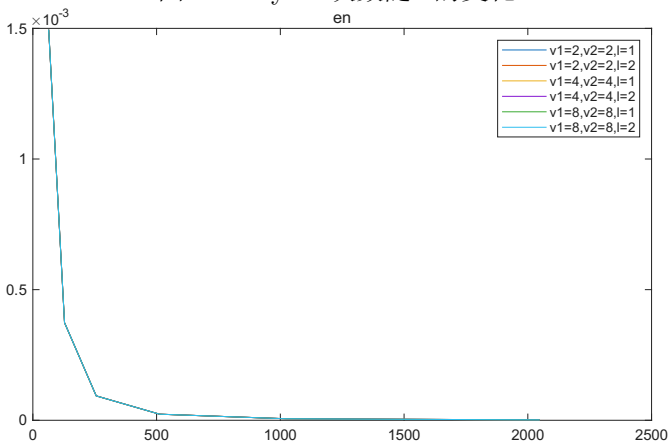

图 8: en随N的变化

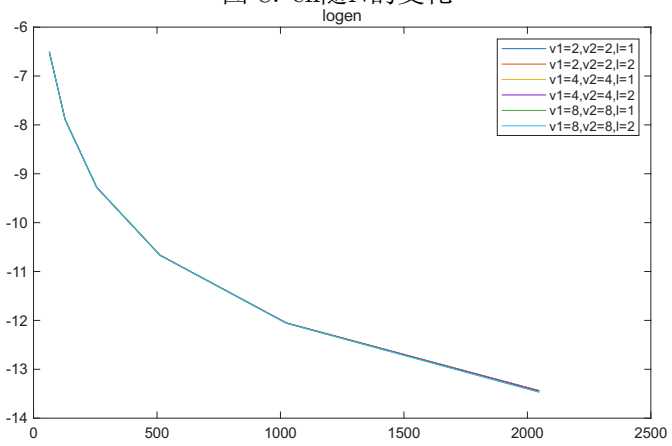

图 9: log(en)随N的变化

可见e_N收敛, 及在本题中不同参数计算速度差距不大, v_1, v_2较大时, V-cycle次数较小
将部分完整数据以表格形式呈现如下

表 2: 不同参数下时间随N的变化

参数	N=64	N=128	N=256	N=512	N=1024	N=2048
$v_1 = 2, v_2 = 2, l = 1$	0.0660	0.1386	0.7367	3.9070	20.3413	159.9633
$v_1 = 2, v_2 = 2, l = 2$	0.0684	0.1458	0.6295	3.5230	22.4530	156.0362
$v_1 = 4, v_2 = 4, l = 1$	0.0494	0.1667	0.7907	3.4477	22.0449	133.9093
$v_1 = 4, v_2 = 4, l = 2$	0.0478	0.1690	0.6362	3.4793	21.4368	133.8237
$v_1 = 8, v_2 = 8, l = 1$	0.0772	0.2732	0.8475	4.8484	28.7869	147.1572
$v_1 = 8, v_2 = 8, l = 2$	0.0716	0.2549	0.9194	4.8578	30.3523	138.8127

表 3: 不同参数下 V-cycle 次数随N的变化

参数	N=64	N=128	N=256	N=512	N=1024	N=2048
$v_1 = 2, v_2 = 2, l = 1$	7	7	7	8	8	8
$v_1 = 2, v_2 = 2, l = 2$	7	7	7	8	8	8
$v_1 = 4, v_2 = 4, l = 1$	5	5	5	5	5	5
$v_1 = 4, v_2 = 4, l = 2$	5	5	5	5	5	5
$v_1 = 8, v_2 = 8, l = 1$	5	5	4	4	4	4
$v_1 = 8, v_2 = 8, l = 2$	5	5	4	4	4	4

3 P2: 使用 Uzawa Iteraion Method

3.1 数值方法

3.1.1 实用共轭梯度法

A对称正定, 解方程

$$Ax = b$$

这等价于求

$$\arg\min_{x\in\mathbb{R}^n} \psi(x) = x^T Ax - 2b^T x$$

step1: 选定初始向量x_0, 下降方向为

$$p_0 = r_0 = b - Ax_0$$

并记

$$\rho_0 = r_0^T r_0, \qquad w_0 = Ap_0$$

取最大下降步长

$$\alpha_0 = \frac{r_0^T r_0}{p_0^T Ap_0} = \frac{\rho_0}{p_0^T w_0}$$

得

$$x_1 = x_0 + \alpha_0 p_0, \qquad r_1 = b - Ax_1 = r_0 - \alpha w_0$$

step(k+1): 在平面

$$\pi = \{x | x = x + k + \xi r_k + \eta p_{k-1}, \epsilon, \eta \in \mathbb{R}\}$$

内找出使ψ下降最快的方向

$$p_k = r_k + \beta_{k-1} p_{k-1}, \qquad w_k = Ap_k$$

为新的下降方向, 其中

$$\beta_{k-1} = -\frac{r_k^T Ap_{k-1}}{p_{k-1}^T Ap_{k-1}} = \frac{r_k^T r_k}{r_{k-1}^T r_{k-1}} = \frac{\rho_k}{\rho_{k-1}}$$

并取最大下降步长

$$\alpha_k = \frac{r_k^T p_k}{p_k^T Ap_k} = \frac{r_k^T r_k}{p_k^T Ap_k} = \frac{\rho_k}{p_k^T w_k}$$

$$x_{k+1} = x_k + \alpha_k p_k, \qquad r_{k+1} = b - Ax_{k+1} = r_k - \alpha_k w_k, \qquad \rho_{k+1} = r_{k+1}^T r_{k+1}$$

共轭梯度法一定收敛到方程的真解, 取停止条件为

$$\sqrt{\rho} < \epsilon \|b\|_2 \qquad or \qquad k \geq k_{max}$$

3.1.2 Uzawa Iteraion Method

求解线性方程

$$\begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} X \\ P \end{pmatrix} = \begin{pmatrix} Y \\ 0 \end{pmatrix}$$

step1: 初值取为$U, V, P = 0$

step2: 精确求解

$$AX_{k+1} = Y - B^T P_k$$

这里采用实用共轭梯度法

step3: 更新压力

$$P_{k+1} = P_k + \alpha(B^T X_k)$$

step4:

$$r_h = \begin{pmatrix} Y \\ 0 \end{pmatrix} - \begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} \hat{X} \\ \hat{P} \end{pmatrix}$$

$$r_0 = Y$$

停止条件为

$$\frac{\|r_h\|_2}{\|r_0\|_2} < 10^{-8}$$

如满足停止条件, 停止迭代, $\hat{X}$为近似解

否则, 回到step2

3.2 具体算法实现

表 4: 辅助函数

文件	作用
p2:/initialize.m	初始化F, G及计算真解
p2:/left_multiply_matrix.m	对矩阵U, V的向量形式左乘$[A, B; B^T, 0]$后化回矩阵形式
p2:/left_multiply_A.m	对矩阵U, V的向量形式左乘A后化回矩阵形式
p2:/left_multiply_B.m	对矩阵P的向量形式左乘B后化回矩阵形式
p2:/left_multiply_BT.m	对矩阵U, V的向量形式左乘B^T后化回矩阵形式
p2:/dot_multiply.m	对两个矩阵, 化成向量形式后求内积

Algorithm 3 Uzawa Iteration Method(p2:/Uzawa.m)

1: $P = 0, r_0 = \sqrt{\|F\|_F^2 + \|G\|_F^2}$
2: **while** $r_h/r_0 > 10^{-8}$**and** $iter < MAXITER$ **do**
3: $[F1, G1]$ =**left_multiply_B**(P, N)
4: $[U, V, iter1]$ =**CG**$(F - F1, G - G1, N, 10^{-8})$
5: $P = P + \alpha$**left_multiply_BT**(U, V, N)
6: $[U1, V1, P1]$ =**left_multiply_matrix**(U, V, P, N)
7: $r_h = \sqrt{\|U1 - F\|_F^2 + \|V1 - G\|_F^2 + \|P1\|_F^2}$
8: **end while**

Algorithm 4 实用共轭梯度法(p2:/CG.m)

$U, V, P = 0, RU = F, RV = G, b = \|[F, G]\|_F, MAXITER = N\sqrt{N}$
while $\sqrt{\rho} > \epsilon b$**and** $iter < MAXITER$ **do**
  $iter = iter + 1$
  if $i == 1$ **then**
    $PU = RU, PV = RV$
    $\rho = \|RU\|_F^2 + \|RV\|_F^2$
  else
    $\hat{\rho} = \rho$
    $\rho = \|RU\|_F^2 + \|RV\|_F^2$
    $\beta = \rho/\hat{\rho}$
    $PU = RU + \beta PU; PV = RV + \beta PV$
  end if
  $[WU, WV]$ =**left_multiply_A**(PU, PV, N);
  $\alpha = \rho/($**dot_multiply**$(WU, PU)+$**dot_multiply**$(WV, PV))$
  $U = U + \alpha PU, V = V + \alpha PV$
  $RU = RU - \alpha WU, RV = RV - \alpha WV$
end while

3.3 数值结果

分别取$N = 64, 128, 256, 512$, 以 Uzawa Iteration Method 求解离散问题, 停机标准为$\|r_h\|_2/\|r_0\|_2 < 10^{-8}$, 并计算误差

$$e_N = h\left(\sum_{j=1}^{N}\sum_{i=1}^{N-1}\left|u_{i,j-\frac{1}{2}} - u(x_i, y_{j-\frac{1}{2}})\right|^2 + \sum_{j=1}^{N-1}\sum_{i=1}^{N}\left|v_{i-\frac{1}{2},j} - v(x_{i-\frac{1}{2}}, y_j)\right|^2\right)$$

其中$x_i = i\frac{1}{N}, y_j = j\frac{1}{N}$, $u(i,j), v(i,j)$分别为计算结果在(x_i, y_j)处的取值

取$\alpha = 1$, 结果如下

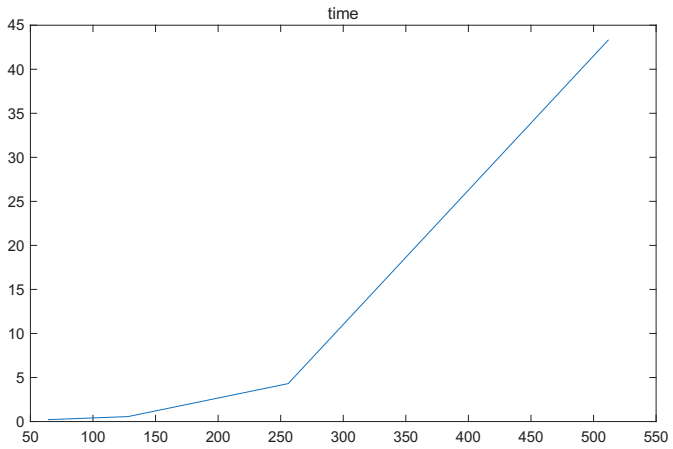

图 10: 时间随N的变化

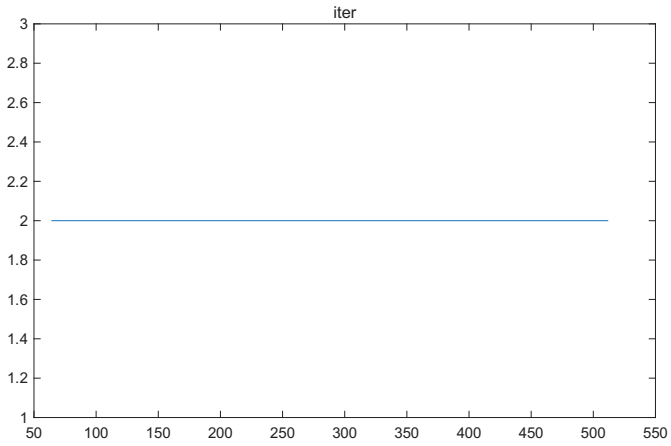

图 11: 外循环次数随N的变化

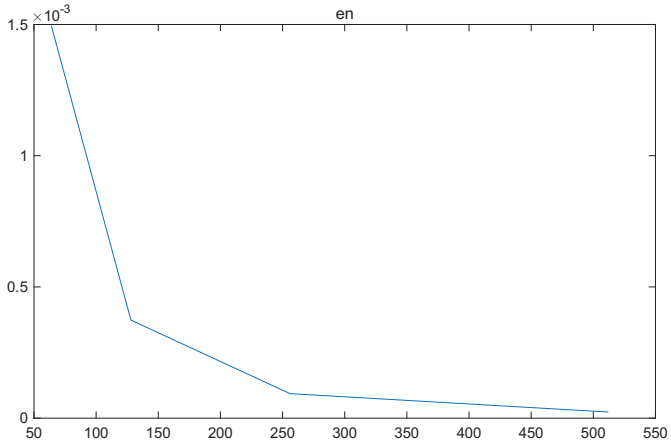

图 12: en随N的变化

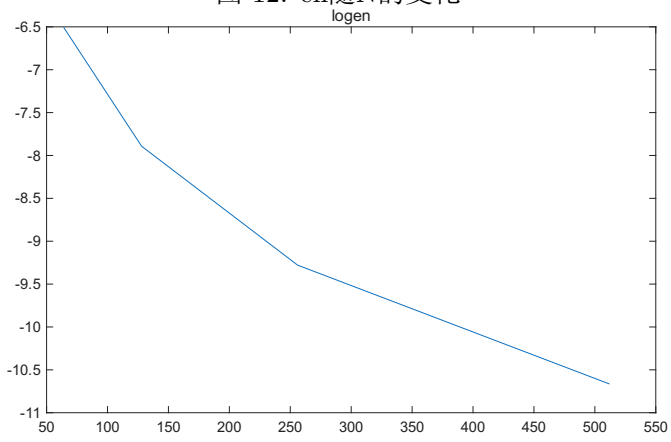

图 13: log(en)随N的变化

4 P3: 使用 Inexact Uzawa Iteration Method

4.1 数值方法

4.1.1 预优共轭梯度法

A对称正定, 解方程

$$Ax = b,$$

共轭梯度法的收敛速度为

$$\|x_k - x_*\|_A \leq 2\left(\frac{\sqrt{\kappa_2}-1}{\sqrt{\kappa_2}+1}\right)^k \|x_0 - x_*\|$$

解原方程等价于解

$$\tilde{A}\tilde{x} = \tilde{b}$$

其中

$$\tilde{A} = C^{-1}AC^{-1}, \qquad \tilde{x} = Cx, \qquad \tilde{b} = C^{-1}b$$

记

$$M = C^2$$

当$\|\tilde{A}\|_2$较小时, 对前述方程使用共轭梯度法求解, 收敛速度较快, 可知M为A的近似时收敛较快

在共轭梯度法的基础上, 给出预优共轭梯度法, 设对于新方程采用共轭梯度法求解时, 得到的各个中间变量为

$$\tilde{x_k}, \tilde{p_k}, \tilde{r_k}, \tilde{\alpha_k}, \tilde{\beta_k}, \tilde{\rho_k}, \tilde{w_k}$$

step1: 选定初始向量

$$x_0 = C^{-1}\tilde{x_0}$$

$$r_0 = b - Ax_0 = C\tilde{b} - C\tilde{A}C \cdot C^{-1}\tilde{x_0} = C\tilde{r_0}$$

$$p_0 = z_0 = M^{-1}r_0 = C\tilde{p_0}$$

$$\rho_0 = r_0^T z_0 = \tilde{r_0}^T \tilde{r_0} = \tilde{\rho_0}$$

step(k+2): 在求解过程中

$$w_k = Ap_k = C\tilde{A}\tilde{p_k}$$

$$\alpha_k = \frac{\rho_k}{p_k^T w_k} = \frac{\tilde{r_k}^T \tilde{r_k}}{\tilde{p_k}^T \tilde{A}\tilde{p_k}} = \tilde{\alpha_k}$$

$$x_{k+1} = x_k + \alpha_k p_k = C^{-1}\tilde{x_{k+1}}$$

$$r_{k+1} = r_k - \alpha_k w_k = C\tilde{r_{k+1}}$$

$$z_{k+1} = M^{-1}r_{k+1} = C^{-1}r_{k+1} = C\tilde{r_{k+1}}$$

$$\rho_{k+1} = r_{k+1}^T z_{k+1} = \tilde{r_{k+1}}^T \tilde{r_{k+1}} = \tilde{\rho_{k+1}}$$

$$\beta_k = \frac{\rho_{k+1}}{\rho_k} = \tilde{\beta_k}$$

$$p_{k+1} = z_{k+1} + \beta_k p_k = C^{-1}\tilde{p_{k+1}}$$

对新方程使用共轭梯度法, 收敛到其真解$\tilde{x_*}$

所以上述预优共轭梯度法收敛到原方程的真解x_*

取停止条件为

$$\sqrt{\rho} < \epsilon\|b\|_2 \qquad or \qquad k \geq k_{max}$$

4.1.2 Inexact Uzawa Iteraion Method

求解线性方程

$$\begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} X \\ P \end{pmatrix} = \begin{pmatrix} Y \\ 0 \end{pmatrix}$$

step1: 初值取为$U, V, P = 0$

step2: 采取预优共轭梯度法, 近似求解

$$AX_{k+1} = Y - B^T P_k$$

在本问题中, 使用 V-cycle 多重网格方法为预条件子

即在采用预优共轭梯度法时, 在计算

$$Mz = r$$

的过程中, 采用 V-cycle 法近似求解

$$Az = r$$

step3: 更新压力

$$P_{k+1} = P_k + \alpha(B^T X_k)$$

step4:

$$r_h = \begin{pmatrix} Y \\ 0 \end{pmatrix} - \begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} \hat{X} \\ \hat{P} \end{pmatrix}$$

$$r_0 = Y$$

停止条件为

$$\frac{\|r_h\|_2}{\|r_0\|_2} < 10^{-8}$$

如满足停止条件, 停止迭代, $\hat{X}$为近似解

否则, 回到step2

4.2 具体算法实现

表 5: 辅助函数

文件	作用
p3:/initialize.m	初始化F, G及计算真解
p3:/left_multiply_matrix.m	对矩阵U, V的向量形式左乘$[A, B; B^T, 0]$后化回矩阵形式
p3:/left_multiply_A.m	对矩阵U, V的向量形式左乘A后化回矩阵形式
p3:/left_multiply_B.m	对矩阵P的向量形式左乘B后化回矩阵形式
p3:/left_multiply_BT.m	对矩阵U, V的向量形式左乘B^T后化回矩阵形式
p3:/dot_multiply.m	对两个矩阵, 化成向量形式后求内积
p3:/limit.m	对F, G应用限制算子, 保持其矩阵形式
p3:/lift.m	对U, V应用提升算子, 保持其矩阵形式
p3:/GS.m	对$AX = Y$做Gauss-Seidel迭代

Algorithm 5 V-cycle 多重网格法(p3:/V_cycle.m)

$U, V = 0, iter1 = log_2(N) - log_2(L)$
while $r_h/r_0 > 10^{-8}$ **and** $iter < MAXITER$ **do**
  $iter = iter + 1$
  for $i = 1 : iter1$ **do**
    for $k = 1 : v_1$ **do**
      $[U, V]$ =**GS**(U, V, F, G, N)
    end for
    $store\ U, V, F, G, N$
    $[U1, V1]$ =**left_multiply_A**(U, V, N)
    $[F, G]$ =**limit**$(F - U1, G - V1, N)$
    $N = N/2$
  end for
  $solve\ U, V, F, G \longrightarrow U, V$
  for $i = iter1 : -1 : 1$ **do**
    $[U, V]$ =**lift**(U, V, N)
    $N = N * 2$
    $U, V = store + U, V$
    for $k = 1 : v_2$ **do**
      $[U, V]$ =**GS**$(U, V, Fstore, Gstore, N)$
    end for
    if $i == 1$ **then**
      $store\ U, V$
    end if
  end for
end while

Algorithm 6 预优共轭梯度法(p3:/PCG.m)

$U, V, P = 0, RU = F, RV = G, b = \|[F, G]\|_F, MAXITER = N$
while $\sqrt{\rho} > \epsilon b(\epsilon = 10^{-8})$**and** $iter < MAXITER$ **do**
  $iter = iter + 1$
  $[ZU, ZV]$ =**V_cycle**(RU, RV, v_1, v_2, N)
  if $i == 1$ **then**
    $PU = ZU, PV = ZV$
    ρ =**dot_multiply**(RU, ZU)+ **dot_multiply**(RV, ZV)
  else
    $\hat{\rho} = \rho$
    ρ =**dot_multiply**(RU, ZU)+ **dot_multiply**(RV, ZV)
    $\beta = \rho/\hat{\rho}$
    $PU = ZU + \beta PU, PV = ZV + \beta PV$
  end if
  $[WU, WV]$ =**left_multiply_A**(PU, PV, N);
  $\alpha = \rho/($**dot_multiply**(WU, PU)+**dot_multiply**$(WV, PV))$
  $U = U + \alpha PU, V = V + \alpha PV$
  $RU = RU - \alpha WU, RV = RV - \alpha WV$
end while

4.3 数值结果及相应分析

分别取$N = 64, 128, 256, 512, 1024, 2048,$ 以 Inexact Uzawa Iteration Method 求解离散问题, 停机标准为$\|r_h\|_2/\|r_0\|_2 < 10^{-8}$, 其中以 V-cycle 多重网格方法为预条件子, 利用共轭梯度法求解每一步的子问题$\mathbf{AU_{k+1}} = \mathbf{F} - \mathbf{BP_k}$, 对不同的$\alpha, \tau, 1, 2, L,$ 比较外循环的迭代次数和 CPU 时间, 并计算误差

$$e_N = h\left(\sum_{j=1}^{N}\sum_{i=1}^{N-1}\left|u_{i,j-\frac{1}{2}} - u(x_i, y_{j-\frac{1}{2}})\right|^2 + \sum_{j=1}^{N-1}\sum_{i=1}^{N}\left|v_{i-\frac{1}{2},j} - v(x_{i-\frac{1}{2}}, y_j)\right|^2\right)$$

其中$x_i = i\frac{1}{N}, y_j = j\frac{1}{N}$, $u(i,j), v(i,j)$分别为计算结果在(x_i, y_j)处的取值
结果如下

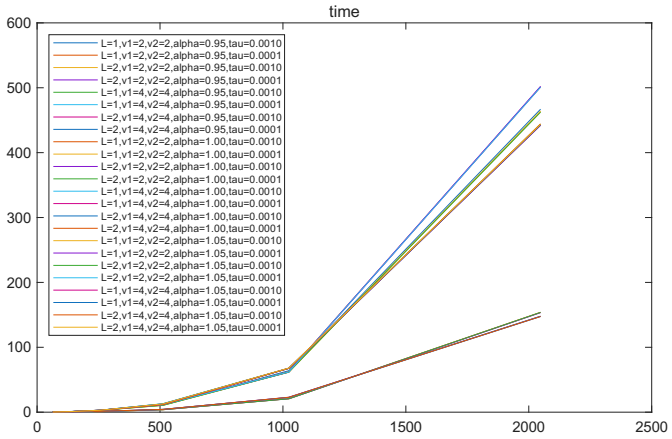

图 14: 不同参数下时间随N的变化

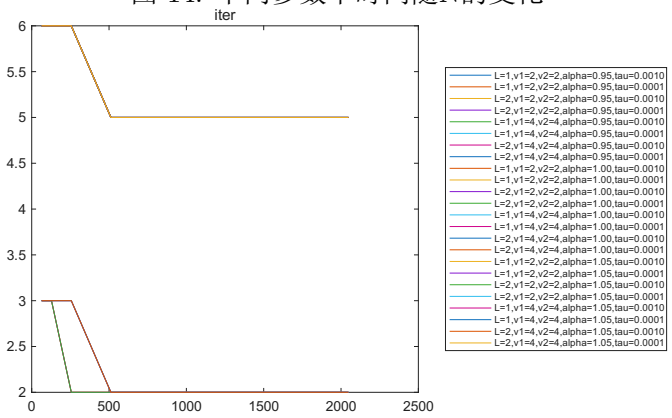

图 15: 不同参数下外循环次数随N的变化

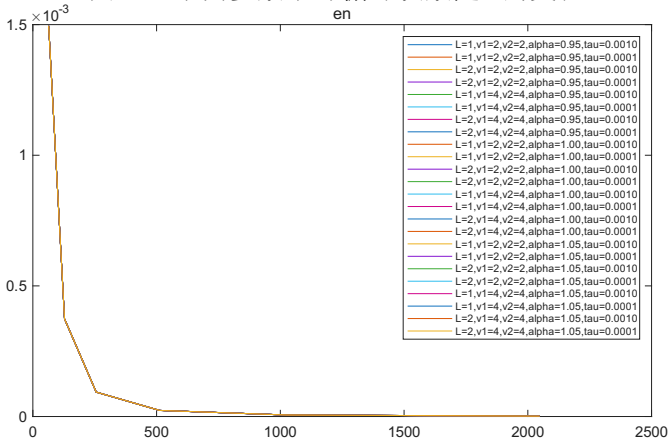

图 16: 不同参数下en随N的变化

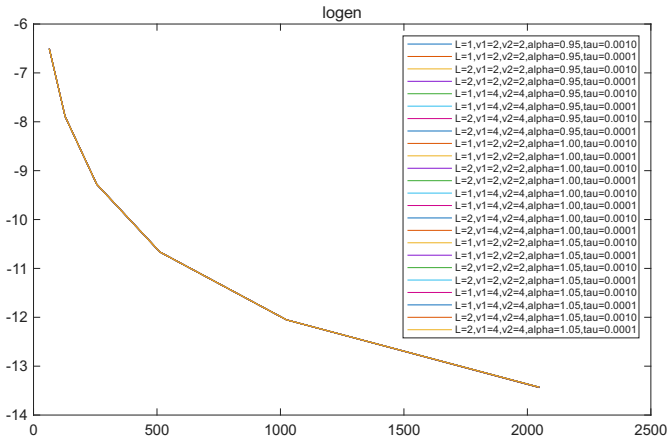

图 17: 不同参数下log(en)随N的变化

可见e_N收敛, $\alpha = 1$时收敛速度显著快于$\alpha = 0.95 or 1.05$, 且迭代次数更小, 而其余参数对结果影响不大
将部分数据用表格形式完整呈现如下

表 6: 不同参数下时间随N的变化

参数	N=64	N=128	N=256	N=512	N=1024	N=2048
$L=1, v_1=2, v_2=2, \alpha=0.95, \tau=0.0010$	0.6831	0.9120	2.9944	10.9916	63.9492	466.9588
$L=1, v_1=2, v_2=2, \alpha=0.95, \tau=0.0001$	0.2831	0.7416	3.0426	11.5018	62.4431	501.3319
$L=2, v_1=2, v_2=2, \alpha=0.95, \tau=0.0010$	0.2783	0.7028	2.7757	10.7246	61.6444	464.2096
$L=2, v_1=2, v_2=2, \alpha=0.95, \tau=0.0001$	0.2646	0.7813	3.0126	11.3898	62.1861	502.1172
$L=1, v_1=4, v_2=4, \alpha=0.95, \tau=0.0010$	0.2721	0.6915	2.7541	11.9746	67.7576	443.8745
$L=1, v_1=4, v_2=4, \alpha=0.95, \tau=0.0001$	0.2989	0.7732	2.9550	13.0916	67.5270	442.6019
$L=2, v_1=4, v_2=4, \alpha=0.95, \tau=0.0010$	0.2535	0.7392	2.7973	11.7912	67.7846	442.1117
$L=2, v_1=4, v_2=4, \alpha=0.95, \tau=0.0001$	0.2854	0.8095	3.0190	12.7961	67.7756	442.6326
$L=1, v_1=2, v_2=2, \alpha=1.00, \tau=0.0010$	0.1061	0.2705	0.7864	3.8139	20.8214	153.8682
$L=1, v_1=2, v_2=2, \alpha=1.00, \tau=0.0001$	0.1284	0.3217	0.8116	3.8405	20.7670	153.7367
$L=2, v_1=2, v_2=2, \alpha=1.00, \tau=0.0010$	0.1019	0.2776	0.8051	3.9084	20.6553	153.4189
$L=2, v_1=2, v_2=2, \alpha=1.00, \tau=0.0001$	0.1265	0.3093	0.8189	3.8054	20.6308	153.8722
$L=1, v_1=4, v_2=4, \alpha=1.00, \tau=0.0010$	0.1035	0.2684	1.0926	4.3201	22.7530	147.3751
$L=1, v_1=4, v_2=4, \alpha=1.00, \tau=0.0001$	0.1187	0.3329	1.2836	4.2956	23.0500	147.6945
$L=2, v_1=4, v_2=4, \alpha=1.00, \tau=0.0010$	0.1123	0.2875	1.1047	4.2912	22.8108	147.9644
$L=2, v_1=4, v_2=4, \alpha=1.00, \tau=0.0001$	0.1234	0.3295	1.2970	4.3109	22.8637	147.5623
$L=1, v_1=2, v_2=2, \alpha=1.05, \tau=0.0010$	0.3170	0.7805	3.0682	10.6040	62.0672	463.2114
$L=1, v_1=2, v_2=2, \alpha=1.05, \tau=0.0001$	0.3073	0.7054	2.9982	11.4006	61.7156	501.4783
$L=2, v_1=2, v_2=2, \alpha=1.05, \tau=0.0010$	0.2919	0.7097	2.7750	10.5026	61.7900	462.4892
$L=2, v_1=2, v_2=2, \alpha=1.05, \tau=0.0001$	0.2706	0.7195	2.9506	11.4420	62.0211	500.9380
$L=1, v_1=4, v_2=4, \alpha=1.05, \tau=0.0010$	0.2567	0.7130	2.8251	11.8441	67.4739	442.9305
$L=1, v_1=4, v_2=4, \alpha=1.05, \tau=0.0001$	0.3051	0.7647	2.9605	12.8265	67.5050	443.1289
$L=2, v_1=4, v_2=4, \alpha=1.05, \tau=0.0010$	0.2489	0.7205	2.8042	11.7245	67.5906	444.3659
$L=2, v_1=4, v_2=4, \alpha=1.05, \tau=0.0001$	0.2905	0.7561	2.9851	12.8375	68.0027	443.6839

表 7: 不同参数下外循环次数随N的变化

参数	N=64	N=128	N=256	N=512	N=1024	N=2048
$L=1, v_1=2, v_2=2, \alpha=0.95, \tau=0.0010$	6	6	6	5	5	5
$L=1, v_1=2, v_2=2, \alpha=0.95, \tau=0.0001$	6	6	6	5	5	5
$L=2, v_1=2, v_2=2, \alpha=0.95, \tau=0.0010$	6	6	6	5	5	5
$L=2, v_1=2, v_2=2, \alpha=0.95, \tau=0.0001$	6	6	6	5	5	5
$L=1, v_1=4, v_2=4, \alpha=0.95, \tau=0.0010$	6	6	6	5	5	5
$L=1, v_1=4, v_2=4, \alpha=0.95, \tau=0.0001$	6	6	6	5	5	5
$L=2, v_1=4, v_2=4, \alpha=0.95, \tau=0.0010$	6	6	6	5	5	5
$L=2, v_1=4, v_2=4, \alpha=0.95, \tau=0.0001$	6	6	6	5	5	5
$L=1, v_1=2, v_2=2, \alpha=1.00, \tau=0.0010$	3	3	2	2	2	2
$L=1, v_1=2, v_2=2, \alpha=1.00, \tau=0.0001$	3	3	2	2	2	2
$L=2, v_1=2, v_2=2, \alpha=1.00, \tau=0.0010$	3	3	2	2	2	2
$L=2, v_1=2, v_2=2, \alpha=1.00, \tau=0.0001$	3	3	2	2	2	2
$L=1, v_1=4, v_2=4, \alpha=1.00, \tau=0.0010$	3	3	3	2	2	2
$L=1, v_1=4, v_2=4, \alpha=1.00, \tau=0.0001$	3	3	3	2	2	2
$L=2, v_1=4, v_2=4, \alpha=1.00, \tau=0.0010$	3	3	3	2	2	2
$L=2, v_1=4, v_2=4, \alpha=1.00, \tau=0.0001$	3	3	3	2	2	2
$L=1, v_1=2, v_2=2, \alpha=1.05, \tau=0.0010$	6	6	6	5	5	5
$L=1, v_1=2, v_2=2, \alpha=1.05, \tau=0.0001$	6	6	6	5	5	5
$L=2, v_1=2, v_2=2, \alpha=1.05, \tau=0.0010$	6	6	6	5	5	5
$L=2, v_1=2, v_2=2, \alpha=1.05, \tau=0.0001$	6	6	6	5	5	5
$L=1, v_1=4, v_2=4, \alpha=1.05, \tau=0.0010$	6	6	6	5	5	5
$L=1, v_1=4, v_2=4, \alpha=1.05, \tau=0.0001$	6	6	6	5	5	5
$L=2, v_1=4, v_2=4, \alpha=1.05, \tau=0.0010$	6	6	6	5	5	5
$L=2, v_1=4, v_2=4, \alpha=1.05, \tau=0.0001$	6	6	6	5	5	5